Chi-Chang Lee

I am a CS Ph.D. student at University of Maryland, College Park. Previously, I earned both my Bachelor's and Master's degrees in Engineering Science and Computer Science from National Taiwan University, where I worked closely with Prof. Yu Tsao and Prof. Hsin-Min Wang. During that period, my research primarily centered on machine learning for audio and speech modalities. I focused on improving robustness to diverse acoustic environments (ICLR'23 & ASRU'23) and on how such robustness can be continually learned and adapted over time (Interspeech'20 & Interspeech'22).

In addition, from 2023 to 2025, I was honored to collaborate with the Improbable AI Lab at MIT CSAIL, led by Prof. Pulkit Agrawal. My work there focused on Reinforcement Learning for robots under a finite interaction budget, aiming to disentangle heuristic signals from the true task objective while still optimizing real but sparse success outcomes. I contributed to the real-world deployment of locomotion policies (ICRA'24) and to an advanced RL algorithm that achieves consistent gains on 31 robotic tasks (NeurIPS'24).

My current research centers on the embodiment gap between human demonstrations and dexterous robot execution, spanning morphological, visual, and action-space dimensions. I investigated these questions at Origami Robotics with Prof. Yonatan Bisk.

Experience

Research Intern May 2026 – Aug. 2026
Origami Robotics (YC W26), San Francisco, CA
Advisor: Prof. Yonatan Bisk
Investigating embodiment gap across morphological, visual, and action-space dimensions to bridge human demonstrations with dexterous robot execution.

Massachusetts Institute of Technology

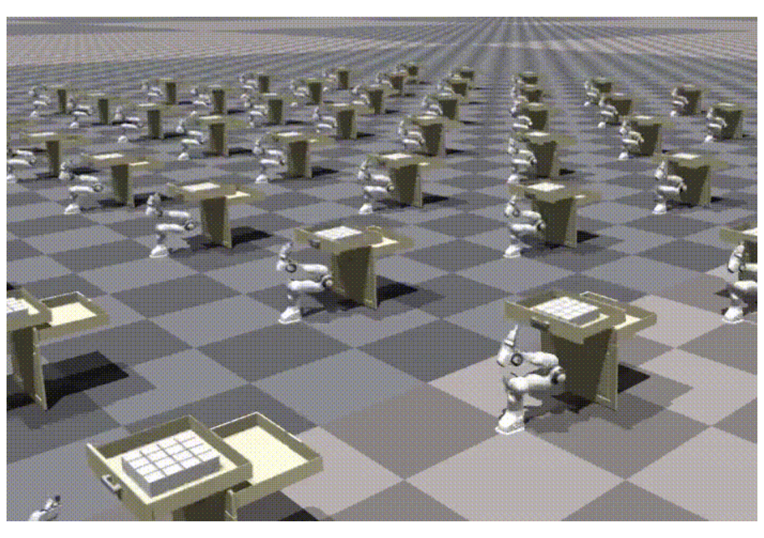
Research Collaborator Jul. 2023 – Sep. 2025
Improbable AI Lab at Massachusetts Institute of Technology
Advisor: Prof. Pulkit Agrawal

Research Assistant Mar. 2019 – Mar. 2024
Bio-ASP Lab at Academia Sinica CITI, Taiwan
Advisor: Prof. Yu Tsao

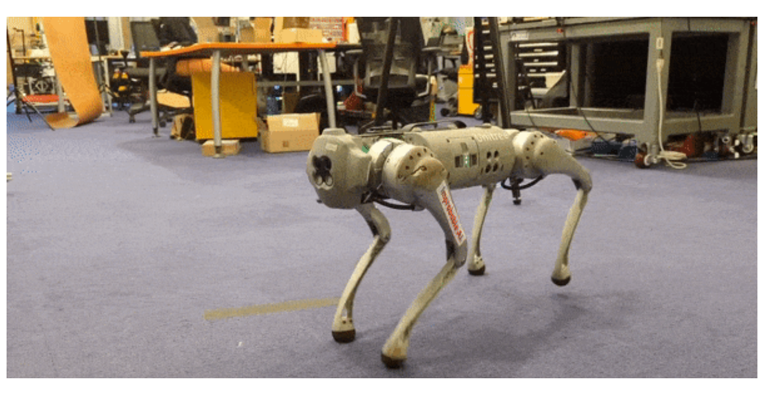
Publications in Robot Learning and Embodied AI

(* indicates equal contribution)

Optimal robot learning under finite interaction budgets - In practice, RL for robotics often struggles with complex environments because it relies on brittle, hand-engineered dense rewards or assumes unrealistic sampling budgets (e.g., relying on proximal heuristics while the true target is a sparse success outcome). I developed a novel RL framework that optimizes for target success outcomes without requiring (1) infinite sampling assumptions or (2) restrictive reward shaping.



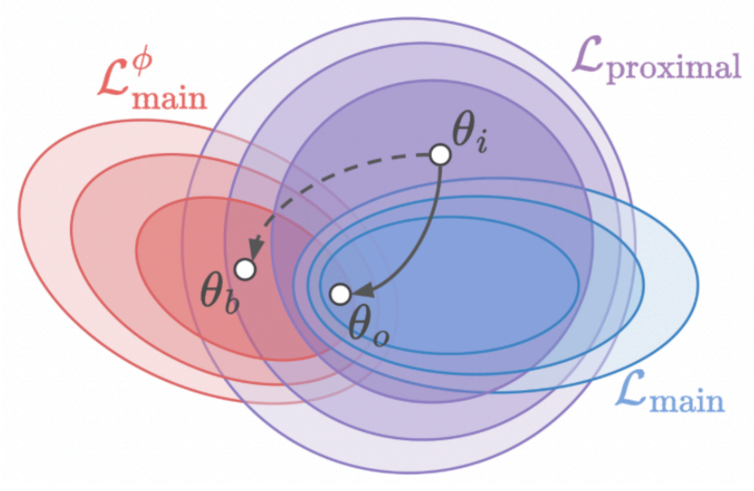
Going Beyond Heuristics by Imposing Policy Improvement as a Constraint
Chi-Chang Lee*, *Zhang-Wei Hong**, *Pulkit Agrawal*
Conference on Neural Information Processing Systems (NeurIPS), 2024
Paper | OpenReview | Video | Code



Maximizing Velocity by Minimizing Energy
*Srinath Mahankali**, ***Chi-Chang Lee****, *Gabriel B. Margolis, Zhang-Wei Hong, Pulkit Agrawal*
International Conference on Robotics and Automation (ICRA), 2024
Paper | Website | Code

Publications in Machine Learning for Audio Modalities

Machine Learning for Audio Modalities - My previous research mainly focused on "normalizing" speech from diverse acoustic conditions for downstream tasks such as Speech-to-Text and Speaker Identification. My related publications cover two main directions: (1) Leveraging multiple label modalities for auxiliary learning and data reuse, by combining synthetic and real-world audio, for example using pairs of noisy and clean speech (synthetic) and pairs of noisy speech and text labels (real-world) to improve robustness. (2) Fast adaptation and continual learning for acoustic normalization in order to mitigate out-of-domain degradation. Given the complexity of speech and acoustic environments, it is unrealistic to assume that audio models will generalize sufficiently on their own, so my work focuses on how to automatically acquire informative data and rapidly recalibrate these models.


$\mathcal{L}^{\phi}_{\text{main}}$
$\mathcal{L}_{\text{proximal}}$
θ_i
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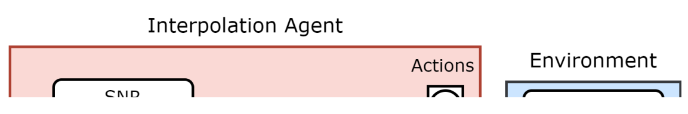
θ_o
$\mathcal{L}_{\text{main}}$

D4AM: A General Denoising Framework for Downstream Acoustic Models
Chi-Chang Lee, *Yu Tsao, Hsin-Min Wang, Chu-Song Chen*
International Conference on Learning Representations (ICLR), 2023
Paper | Code



LC4SV: A Denoising Framework Learning to Compensate for Unseen Speaker

Updated VLM/LLM side projects in 2025

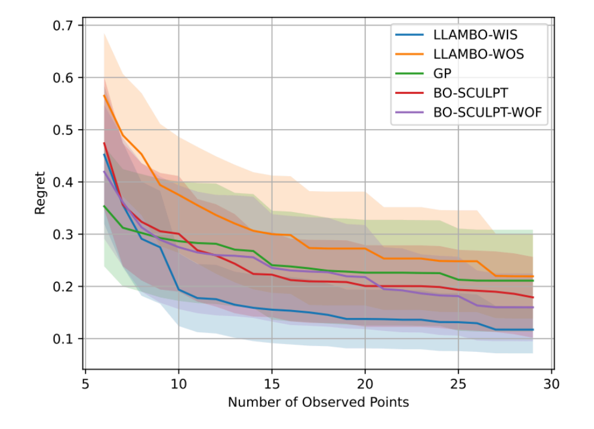
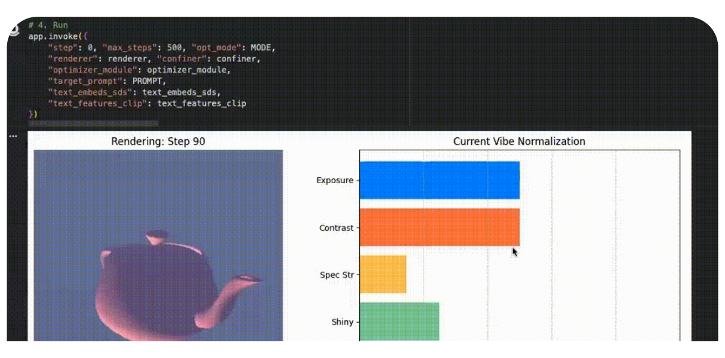


Figure 1: Performance of BO-SCULPT evaluated on the Bayesmark subset.

In-Context Bayesian Optimization incorporating improvement constraints

This is a 14-day side project in 2025 that aims to expand in-context learning for Bayesian optimization by introducing a reference policy and a policy-improvement view. I target two goals: (1) acquiring improvement / convergence support via constrained updates relative to the reference policy, and (2) replacing domain-specific task descriptions (e.g., dataset/model info in hyperparameter tuning) with task-agnostic statistical signals produced by the reference policy (e.g., mean/uncertainty summaries), improving scalability to unseen BO tasks.
Paper | Code

Text-to-Appearance Inverse Rendering with Vision–Language Feedback



```
# 4. Run
app.invoke({
    "step": 0, "max_steps": 500, "opt_mode": MODE,
    "renderer": renderer, "confiner": confiner,
    "optimizer_module": optimizer_module,
    "target_prompt": PROMPT,
    "text_embeds_sds": text_embeds_sds,
    "text_features_clip": text_features_clip
})
```

We study unpaired text-to-appearance inverse rendering, where a differentiable renderer is optimized using only text prompts, without any reference images. Since direct CLIP-based optimization from text is often vague and under-constrained, we introduce vision–language feedback as an optimization confiner. The VLM (1) augments ambiguous textual descriptions into concrete, parameter-aware instructions and (2) determines which scene

Teaching

Teaching Assistant, Machine Learning, National Taiwan University, Taiwan 2021 Fall

Teaching Assistant, Time Frequency Analysis and Wavelet Transforms, National Taiwan University, Taiwan 2018 Fall

Honor

Second Place, IC/CAD Contest 2019

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